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DBMS Assignment - (2)

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CSE-D

① What is normalisation? Explain 2NF and 3NF with example.

Ans) Normalisation:- Analyzing the given relation schemas based on their FDC and primary keys to achieve the desirable properties of

(i) minimizing redundancy and

(ii) minimizing the insertion, deletion & update anomalies.

2NF:- A relation R is in 2NF if it is in 1NF & every non-prime attribute in R is fully functionally dependent on the primary key of R.

Ex:- Table - Purchase - Detail

Customer ID	State ID	Purchase Location
1	1	Los Angeles
1	3	San Francisco
2	1	Los Angeles
3	2	New York
4	3	San Francisco

→ This relation is in 1NF.

→ This table has a composite primary key { Customer ID, State ID }.

→ The non-key attribute is { Purchase Location }.

⇒ $\{ \text{Customer ID}, \text{Store ID} \} \rightarrow \{ \text{Purchase Location} \}$ is a partial dependency, because

$\{ \text{Store ID} \} \rightarrow \{ \text{Purchase Location} \}$

⇒ That means $\{ \text{Purchase Location} \}$ only depends on $\{ \text{Store ID} \}$, which is only part of the primary key.

⇒ Therefore, this table is not in second normal form.

⇒ So, decompose the relation into two relations.

⇒ In one relation, take attributes which are in partial dependency (part of candidate key $\rightarrow$ non-key attribute)
 $R_1 (\text{Store ID}, \text{Purchase Location})$

⇒ In other relation, take remaining attributes from the original relation (Customer ID) and one common attribute (Store ID) between the decomposed relations.

$R_2 (\text{Customer ID}, \text{Store ID})$

Table - Store

R_1	Store ID	Purchase Location
	1	LA (Los Angeles)
	2	New York
	3	San Francisco

Table - Purchase

R_2	Customer ID	Store ID
1	→	1
1	→	3
2	→	1
3	→	2
4	→	3

3NF:- A relation R is in 3NF if it is in 2NF and no non-prime attributes of R is transitively dependent on primary key.

Ex:- Table - Book - Detail

Book ID	Genre ID	Genre Type	Price
1	1	Gardening	25.99
2	2	Sports	14.99
3	1	Gardening	10.00
4	3	Travel	12.99
5	2	Sports	17.99

⇒ The relation is in 2NF.

⇒ Book ID → Genre ID and

Genre ID → Genre Type (non-prime attribute)

Therefore, Book ID → Genre Type

⇒ This relation has transitive dependency. So this relation is not in 3NF.

⇒ Now decompose the relation into the following relations:

Table - Genre

R1

Genre ID	Genre Type
1	Gardening
2	Sports
3	Travel

Table - Book

R2

Book ID	Genre ID	Price
1	1	25.99
2	2	14.99
3	1	10.00
4	3	12.99
5	2	17.99

② Consider a relation $R(ABCDEF)$ with FD's = $\{A \rightarrow BC, D \rightarrow E$ and $AD \rightarrow F$. Decompose the relation upto BCNF.

Sol:- RHS of FD's, A and D are missing

AD is a candidate key

1NF:- This relation has a candidate key, so the given relation is in 1NF.

2NF:- $A \rightarrow BC$, B & C are non-prime attributes which are partially depending on the A.

So relation is not in 2NF

$$A^+ = \{ABC\}$$

⇒ Decompose the relation into:-

$R_1(ABC)$

$R_2(ADEF)$

⇒ R_1 is in 2NF

⇒ $R_2(ADEF)$

$AD \rightarrow F$

$D \rightarrow E$

$$AD^+ = \{ADEF\}$$

$D \rightarrow E$ is a partial dependency.

So R_2 is not in 2NF

$$D^+ = \{DE\}$$

decompose the relation R_2 into R_{21} & R_{22} :-

(3)

$R_1(ABC), R_2(DE), R_3(DAF)$

R_1 & R_2 are in 2NF.

3NF: - Consider the 2NF relations

$R_1(ABC), R_2(DE), R_3(DAF)$

Consider $R_1(ABC), A \rightarrow BC \Rightarrow$ it is not transitive dependency.

So R_1 is in 3NF.

$R_2(DE) D \rightarrow E \Rightarrow$ it is not transitive dependency.

So R_2 is in 3NF.

$R_3(DAF) AD \rightarrow F \Rightarrow$ it is not transitive dependency.

So R_3 is in 3NF.

BCNF: - Consider 3NF relations.

$R_1(ABC), R_2(DE), R_3(DAF)$

Given

$F.D \Rightarrow A \rightarrow BC$

$D \rightarrow E$

$AD \rightarrow F$

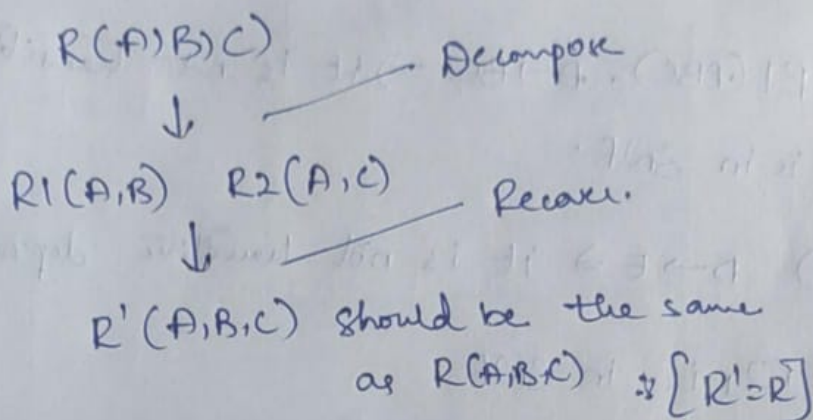
$A \rightarrow BC$:- A is a superkey. So R_1 is in BCNF.

$D \rightarrow E$:- D is a superkey. So R_2 is in BCNF.

$AD \rightarrow F$:- AD is a superkey. So R_3 is in BCNF.

(3) Define the following:-

① Lossless join decomposition:- A decomposition is lossless if we can recover original relation from the decomposed relations.



② Dependency preserving decomposition:- All functional dependencies are preserved in the decomposed relations.

③ Data Redundancy:- Unnecessary repetition of data in a database leading to wastage of space and potential inconsistencies.

④ What is serializability? Explain Conflict serializability with an example?

Serializability:- Ensures that concurrent transactions produce the same result as some serial execution.

Conflict Serializability:- If a schedule can be transformed into a serial schedule by swapping non-conflicting operations.

Ex 1			
T ₁	T ₂	T ₁	T ₂
read(A)		read(A)	
write(A)		write(A)	
	read(A)		
	write(A)		
read(B)		read(B)	
write(B)		write(B)	
	read(B)		read(A)
	write(B)		write(A)
			read(B)
			write(B)

⑤ Explain Timestamp based protocol.

- Each transaction is issued a timestamp when it enters the system.
- If an old transaction T_i has time-stamp $TS(T_i)$, a new transaction T_j is assigned time-stamp $TS(T_j)$ such that $TS(T_i) < TS(T_j)$.
- Time-stamps determine the serializability order.

W-Timestamp(Q) :- is the largest timestamp of any transaction that executed write(Q) successfully.

R-Timestamp(Q) :- is the largest time-stamp of any transaction that executed read(Q) successfully.

⑥ Explain any one approach in log based recovery.

Deferred Database Modification

- Updates are not written to DB until transaction commits.
- If crash happens before commit → no changes made.
- After commit → log is used to update DB.

⑦ Explain different types of indices.

→ Primary Index:- On primary key basis of the table, if the index is created then it is known as primary indexing.

→ Dense Index:- Contains an index record for every search key value in the data file. It makes searching faster.

→ Spase Index:- In the data file, index record appears only for a few items in the data file. Each item points to a block.

→ Clustering Index:- Clustered index is defined on an ordered data file. The data file is ordered on a non-key field.

→ Secondary Index:- Secondary index may be generated from a field which is a candidate key and has a unique value in every record, or a non-key with duplicate values.

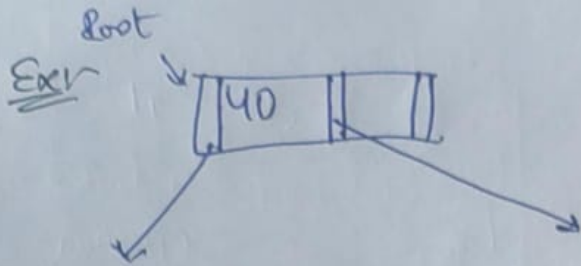
⑧ Explain ISAM with an example.

ISAM (Indexed Sequential Access Method):-

Index built on sorted data.

→ Static Structure.

→ Uses a main index, and overflow areas for updates.



⑨ Explain B+ tree with example.

- ⇒ Dynamic structure - can be updated without using overflow pages.
- ⇒ All data in leaf nodes.
- ⇒ Balanced tree index
- ⇒ Internal nodes only store keys for navigation.

Ex: -

